Mechanical characterization of hollow PVC–ground coffee husk composite elements for load-bearing housing applications

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Highlights

- Standardized mechanical characterization of hollow polymer–fiber housing elements.
- Adapted ASTM procedures enable testing of hollow extruded structural profiles.
- Composite walls exhibit compression- and shear-dominated structural capacity.
- Wall and deck elements show comparable strength despite different geometries.
- Strength levels align with published composites and conventional wood materials.

Abstract

Polyvinyl Chloride-based Natural Fiber Reinforced Plastic Composites (PVC-NFRPCs) are increasingly used in construction; however, standardized mechanical characterization of hollow extruded products suitable for structural design remains limited, constraining their adoption in load-bearing housing components. This study applies and extends the ASTM D7031 characterization framework through practical adaptations required for hollow extruded PVC–coffee husk NFRPC wall and deck elements and a PVC-only channel used in a load-bearing housing system. The results show that wall products exhibit compressive, shear and tensile strength that are direction-dependent due to particle orientation and local failure modes. Deck products displayed mechanical performance comparable to wall

elements, with differences within ±10 % in strength and ±15 % in stiffness, whereas the channels exhibited higher tensile strength but lower stiffness. The measured strengths are consistent with values reported for PVC-based composites with alternative fillers and fall within the performance range of conventional structural materials, supporting the reproducibility of the proposed protocol and the applicability of PVC-NFRPCs as application-tailored construction materials for housing systems.

## Introduction

In 2021, the World Bank estimated that 33 % of rural households in 64 emerging economies lived in inadequate dwellings [1]. A key factor behind the inefficiency of governments and markets to address this issue is the difficulty of supplying construction materials to remote regions. Conventional materials such as cement, steel, and masonry pose challenges due to their considerable weight, which complicates transportation and raises costs, while also contributing to carbon emissions [2]. In this context, novel structural materials that are lightweight, durable, and environmentally sustainable provide effective alternatives. Among these, natural fiber reinforced plastic composites (NFRPCs) stand out as one promising solution to address rural housing demand while supporting sustainability goals.

NFRPCs are synthetic materials, typically molded into hollow, thin-walled profiles for optimized strength-and-stiffness-to-weight ratios. NFRPCs are composed of a polymeric matrix, typically obtained from recycled plastic, natural fiber fillers, and complementary additives [3], [4]. In construction, such plastic composites have been successfully introduced as nonstructural architectural components, including exterior decking and cladding [5], [6]. Such products reduce maintenance costs, are durable against moisture and insects, and ultimately represent cost-effective life cycle assessments [5]. Composite formulations based on polyvinyl chloride (PVC) have demonstrated superior performance in macro-mechanical strength, UV resistance, dimensional stability, compatibility with fibers, and fire resistance [7], [8] compared to alternatives based on other polymers. These characteristics position PVC-NFRPCs as lightweight, durable, and sustainable candidates for structural applications, particularly in rural housing contexts.

Nonetheless, research on the mechanical evaluation of plastic composites for structural applications remains limited. Some of the earliest and most comprehensive contributions to the mechanical characterization of plastic composites originated at Washington State University, most notably the work of Haiar [9] for their use in marine fender systems. Haiar proposed a set of mechanical testing procedures for plastic composites adapted from existing wood-based standards. While this research established a foundation for characterizing key mechanical properties, it did not address other important mechanical parameters, such as tensile strength or parallel to extrusion shear strength, and its procedure, as an initial proposal, did not correspond to any reviewed and approved standard.

The growing use of plastic composites and the resulting research efforts have driven the development of standardized procedures for their mechanical characterization. Among these, the ASTM D7031–11 *Standard Guide for Evaluating Mechanical and Physical Properties of Wood-Plastic Composite Products* [10] is one of the most comprehensive references, covering a wide range of mechanical properties and explicitly accommodating structural applications. In contrast, other standards, such as ASTM D6662 [11] and ISO 16616:2015 [12], are limited to decking products and assess a limited set of properties, which restricts their application. Although terminology differs, according to the definitions given by ASTM D7031, the term wood-plastic composites indistinctively includes NFRPC. Therefore, ASTM D7031 provides a robust framework for extending the evaluation of NFRPCs from nonstructural uses to structural applications. However, because ASTM D7031 references ASTM D143 [11] for several test procedures originally intended for solid-wood specimens, adaptations are required when applying them to hollow cross-section NFRPCs.

Despite the practical direction of standard development, most recent research on PVC-NFRPCs has focused on evaluating reduced-scale specimens to calibrate theoretical models [12] or to assess the individual influence of blend constituents [3], [7], [13]. While expanding the understanding of NFRPCs, this approach is inadequate to describe their performance as structural members, as it does not evaluate all relevant properties, nor assess possible local-failure limit states. To the author's knowledge, very limited research has comprehensively applied a standardized characterization protocol to a PVC-NFRPC material in a manner that includes the full suite of mechanical properties required for structural design.

The objective of this study is to experimentally evaluate the mechanical performance of hollow extruded PVC-based natural fiber reinforced plastic composite (PVC-NFRPC) elements intended for load-bearing housing applications using a standardized, design-oriented testing framework. This work applies ASTM D7031 as the primary characterization guide and introduces practical adaptations required to assess hollow structural profiles subjected to compression, shear, bending, and tension. To the authors' knowledge, this represents the first comprehensive mechanical characterization of hollow PVC-NFRPC construction elements explicitly oriented toward housing applications, where material behavior is evaluated in relation to structural function, load paths, and design-relevant limit states. By deriving characteristic mechanical properties and comparing the results with published PVC-based composite data and conventional structural materials, the study demonstrates the reproducibility, applicability, and design relevance of an ASTM-based protocol for PVC-NFRPC housing systems. Accordingly, while long-term durability and fire-related considerations—such as ultraviolet exposure, creep and creep-rupture behavior, and fire performance—are critical for structural deployment, the present study is intentionally scoped to short-term mechanical characterization, with these aspects addressed in discussion to help contextualize the results and encourage further research.

## NFRPCs prototype for rural housing

This study evaluates the NFRPCs used in the single-story system developed by Woodpecker S.A.S. in Colombia for rural housing, given its potential as a scalable solution for other emerging economies. The system draws inspiration from the principles of log construction but replaces traditional materials with NFRPCs sourced from recycled materials [14]. The system is composed of three fundamental elements (Fig. 1): the primary element referred to as the "wall" type, and two complementary elements

## Materials

Both wall and deck elements are made from a blend of ground coffee husks combined with recycled unplasticized PVC [15], while the channels are produced only from recycled unplasticized PVC. The NFRPC blend used for the wall and deck elements was initially developed in a previous parametric research program conducted through a collaboration between the manufacturer and the Mechanical Engineering Department at Universidad de Los Andes [16], [17]. While the NFRPC blend design is beyond the

## Mechanical property tests

The testing program was designed to characterize the mechanical strengths of the composite elements (Fig. 2) in direct relation to their governing load paths within the housing prototype (Section 2). Wall, deck, and channel elements were evaluated under compression, shear, bending, and tension in directions both parallel and perpendicular to the extrusion process, reflecting their structural roles under gravity, wind, and uplift demands.

ASTM D7031 was adopted as the primary characterization

## Dimensional analysis

The mechanical characterization protocol required a detailed dimensional record of all tested samples. Table 3 presents the statistical description of the products' geometry and the corresponding lognormal fit parameters, which provided the best representation for most measured properties. The dimensional characterization was based on the measure records from the test samples; thus, the sample size corresponds to the number of repetitions.

Overall, the results in Table 3 indicate that the

Influence of cross-sectional geometry

Differences in cross-sectional geometry between the deck and wall products (in Table 3) had a limited influence on perpendicular shear and flatwise bending (Table 4). Although the net cross-sectional area of the wall elements (28.3 $cm^2$) is approximately 1.5 times that of the deck elements (18.9 $cm^2$), the mean perpendicular shear strengths differed by only about 10 % (±1 MPa). Similarly, despite a ratio of approximately 10.4 between their flatwise moments of inertia, bending strengths differed

Conclusions

This study presented an experimental evaluation of hollow polyvinyl chloride–based natural fiber reinforced plastic composite (PVC-NFRPC) elements intended for load-bearing housing applications. By applying a standardized, application-oriented testing framework to full-scale extruded products, the work addresses a key gap in the mechanical characterization of PVC-based composites, as first step in the evaluation of their structural feasibility. The main conclusions are summarized as follows.

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CRediT authorship contribution statement

**C.S. Oliveros:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **J.F. Correal:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Visualization, Project administration, Resources, Supervision, Funding acquisition. **J.A. Medina:** Writing – original draft, Resources, Supervision

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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